Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/23/2005
PI Name:	Golden, Barbara L. Ph.D.		
Project Title:	Engineering a Ribozyme for Diffraction Properties		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	barbgolden@purdue.edu	**Fax:**	FY 765-494-7897
PI Organization Type:	UNIVERSITY	**Phone:**	765-496-6165
Organization Name:	Purdue University		
PI Address 1:	Department of Biochemistry		
PI Address 2:	1153 Biochemistry Building		
PI Web Page:	http://www.hhmi.org/		
City:	West Lafayette	**State:**	IN
Zip Code:	47907-1153	**Congressional District:**	4
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	09/01/2001	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Monaco, Lisa A	**Contact Phone:**	'256 544-4542
Contact Email:	lisa.a.monaco@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1833		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In vitro evolution will be used to develop ribozymes that are thermostable. The thermostable ribozymes will be crystallized. X-ray diffraction analysis will be used to characterize the diffraction characteristics of the crystals. We hope that this study will result in a detailed view of a ribozyme active site at atomic resolution. Furthermore, we hope to develop a generalized strategy for examining the atomic-resolution of multidomain ribosomes. To develop methods in improving X-ray diffraction from RNA crystals. We are trying to understand the structure and function of self-splicing group I introns at the atomic level using X-ray crystallography. These studies are hampered by the low diffraction limits that are typical of crystals formed by multi-domain RNAs. We wish to develop a general method for ‘designing’ RNAs that are more suited for analysis by X-ray crystallography, allowing a more varied structural database of RNA active sites to be obtained in our lab and in other laboratories.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have solved the X-ray crystal structure of a 242 nucleotide ribozyme derived from a self-splicing group intron. This structure reveals the three-domain architecture of group I introns and provides a picture of the peripheral insertions that are characteristic of introns from phage. The structure of the active site suggests that interactions between the ribozyme and its substrate help to fully activate the intron for catalysis. Comparison of the structure with previous studies suggest that modest conformational changes accompany the self-splicing reaction.
Bibliography Type:	Description: (Last Updated: 10/21/2009)
Articles in Peer-reviewed Journals	Golden BL, Kim H, Chase E. "Crystal structure of a phage Twort group I ribozyme-product complex." Nature Structural and Molecular Biology, 2005 Jan;12(1):82-9. PubMed PMID: 15580277 , Jan-2005
Articles in Peer-reviewed Journals	Chase E, Golden BL. "Crystallization and preliminary diffraction analysis of a group I ribozyme from bacteriophage Twort." Acta Crystallogr Sect F Struct Biol Cryst Commun. 2005 Jan 1;61(Pt 1):71-4. PubMed PMID: 16508095 , Jan-2005